MILITARY SPECIFICATION SHEET

CONNECTOR, ELECTRICAL, INSERT (INSULATOR), MALE RECTANGULAR, POLARIZED, CENTER SCREW LOCK, FOR 56 REMOVABLE CONTACTS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-28731.

INCHES	MM
.005	0.13
.043	1.09
.044	1.12
.075	1.90
.090	2.29
.110	2.79
.125	3.18
.130	3.30
.138	3.51
.140	3.56
.150	3.81
.187	4.75
.234	5.94
.250	6.35
.265	6.73
.400	10.16
.412	10.46
.468	11.89
.681	17.30
.798	20.27
.900	22.96
1.015	25.78
1.140	28.96
1.710	43.43
1.900	48.26
2.280	57.91
2.576	65.43
2.592	65.84
3.187	80.95

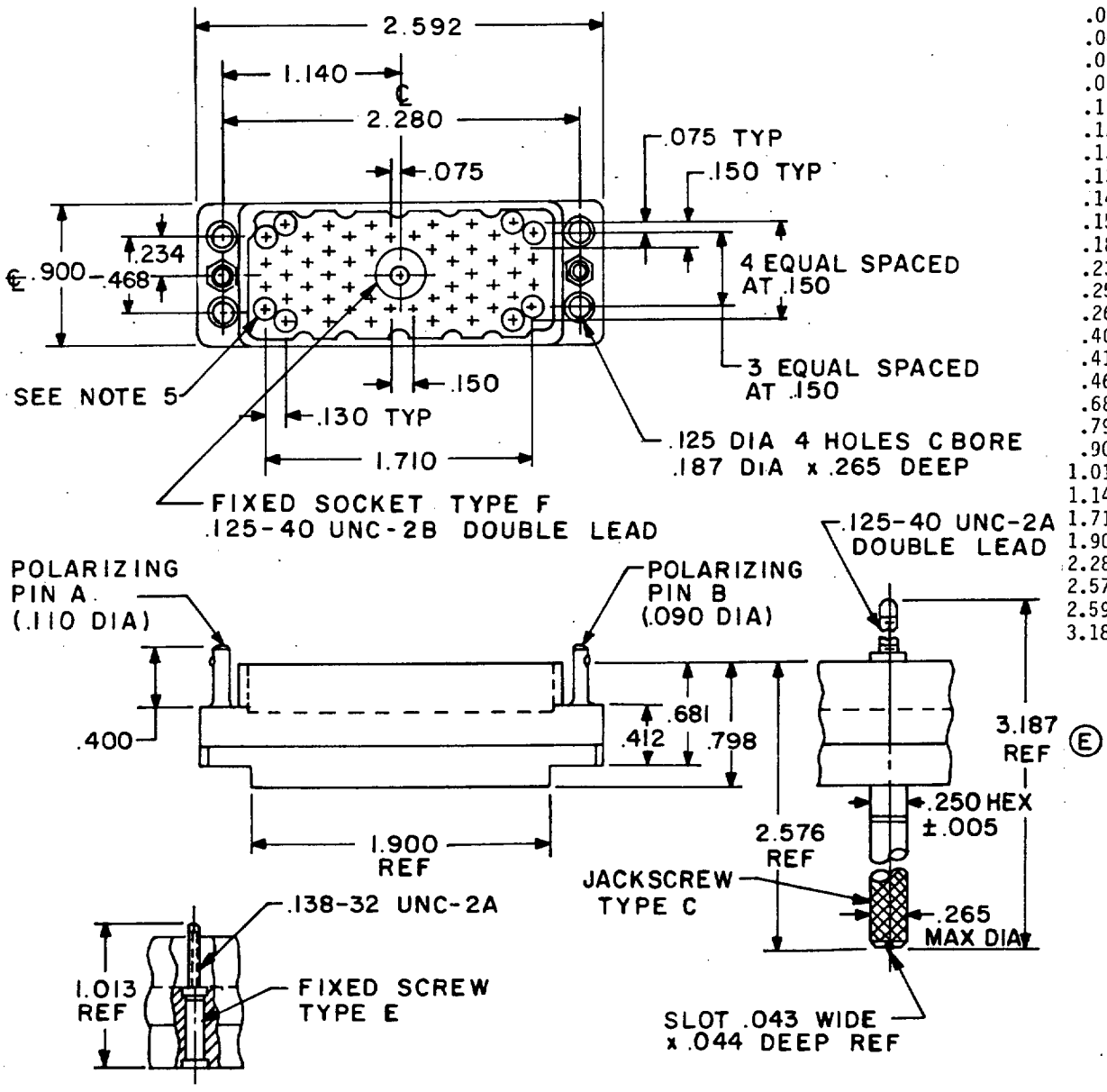
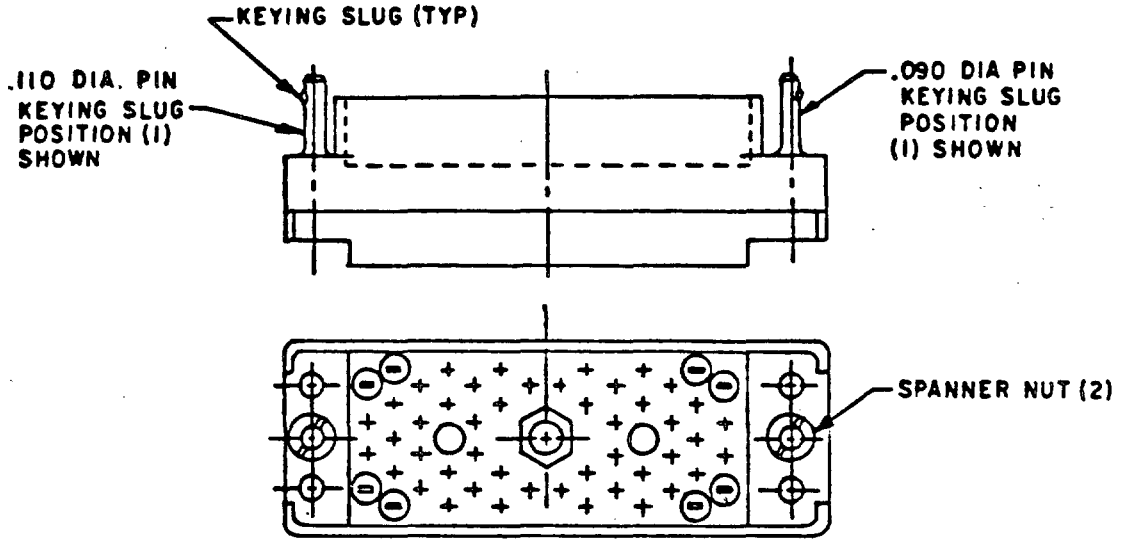


FIGURE 1. Connector, male rectangular.

Ⓔ denotes changes

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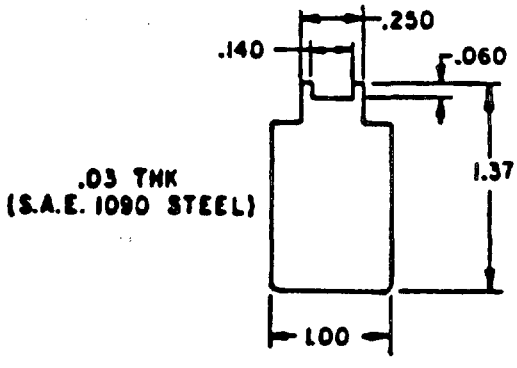
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POLARIZING POSITION CHANGE

Keying position on both polarizing pins can be rotated to desired position by loosening spanner nut (counter-clockwise), resetting pin in one of six positions then retightening spanner nut (clockwise).

NOTE: Overtightening of spanner nut can fracture connector. Spanner keying wrench shown below.



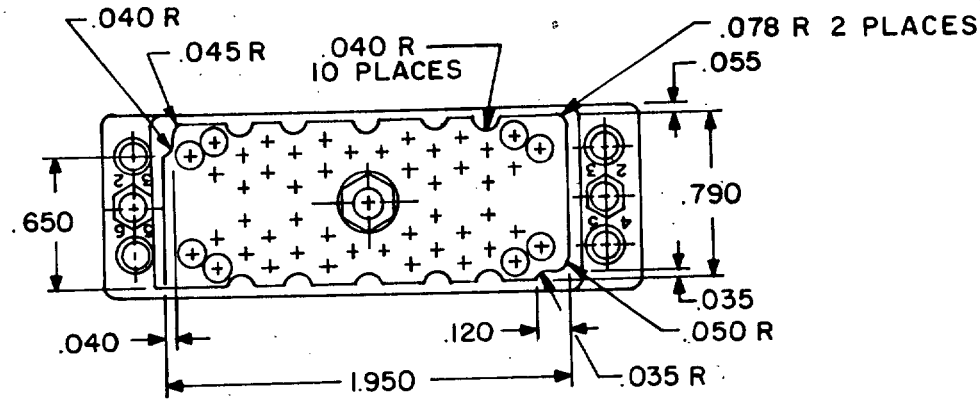
SPANNER WRENCH

INCHES	MM
.03	0.76
.060	1.52
.090	2.29
.110	2.79
.140	3.56
.250	6.35
1.00	25.4
1.37	34.8

Configuration A - Connector polarization guide (layout)

FIGURE 1. Connector, male rectangular - Continued.

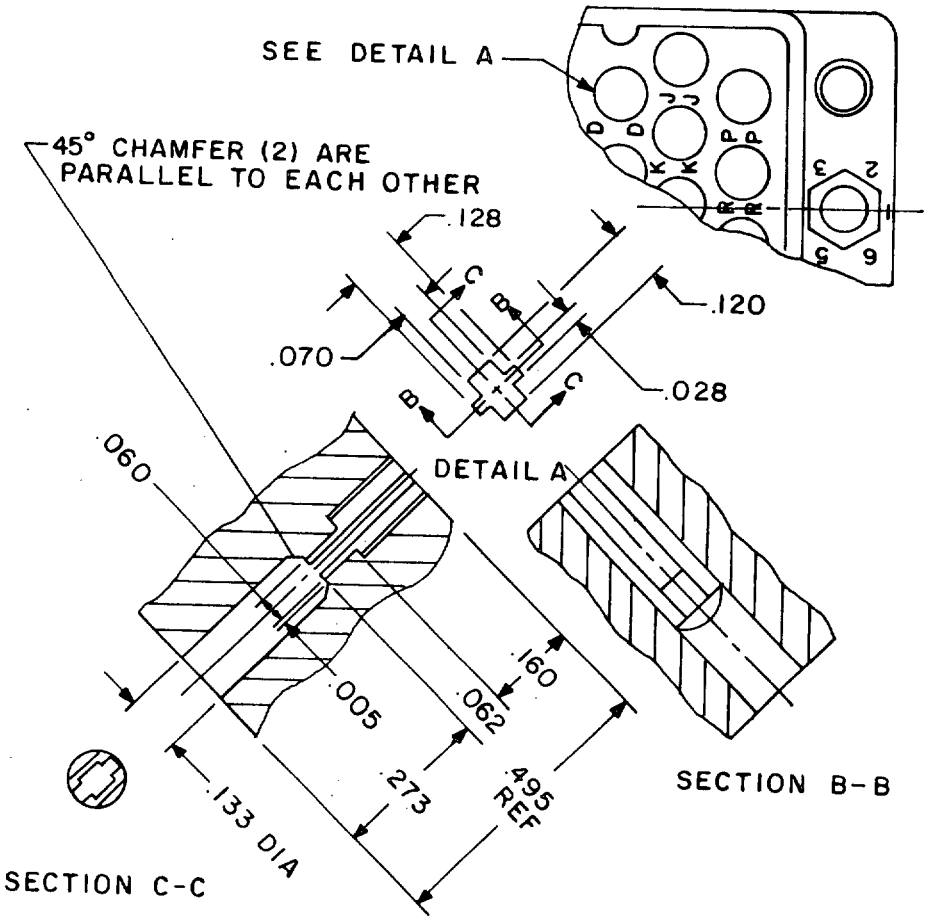
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INCHES	MM
.035	0.89
.040	1.02
.045	1.14
.050	1.27
.055	1.40
.078	1.98
.120	3.05
.650	16.51
.790	20.07
1.950	49.53

FIGURE 1. Connector, male rectangular - Continued.

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INCHES	MM
.005	0.13
.022	0.71
.060	1.52
.062	1.57
.070	1.78
.128	3.25
.133	3.38
.160	4.06
.273	6.93
.495	12.57

Configuration B - Contact hole detail

FIGURE 1. Connector, male rectangular - Continued.

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Tolerances between any two contact centers shall be ±.004 (0.10 mm).
4. Flatness, squareness, and parallelism shall be within dimensional tolerances.
5. Contact hole details shall be in accordance with configuration B.
6. Mating connector information:
   M28731/13-0073 and -0075 and mate with M28731/14-0074, -0076 and -0077.
   M28731/13-0074, -0076 and -0077 mate with M28731/14-0073 and -0075.
   M28731/13-0078, and -0079 and -0080 mate with M28731/14-0078 and -0079.
7. Polarization information:
   Polarization pins are provided for indexing to preclude intermixing of adjacent connectors. By using a standard type spanner wrench, polarization pins can be rotated to desired position. For parts supplied directly to the government, polarization will be in the 1-1 position. For other polarization positions see connector polarization guide configuration A (see marking, note 13d). For direct government acquisition a tool is supplied in the unit package for polarization socket adjustment.
8. Contact and terminal designations:
   Contacts and terminals shall be designated on the insulator using letters in the location shown on figure 1.
9. Tolerance shall be ±.010 (0.25 mm) unless otherwise specified.
10. Patent notice:
    Patent No. 3,248,686 owned by Elco Corporation. The government has a royalty-free license under the above patent for the benefit of manufacturers of items called for in this specification, either for the government or for use in equipment to be delivered to the government.
11. For direct government acquisition crimp contacts M28731/20-0001 are furnished in a plastic bag in each unit package (see marking, note 13a).
12. For direct government acquisition solder hole and wrappost contacts M28731/25-0001 and /26-0001 are factory installed in connector body (see marking, note 13a).
13. Marking:
    a. Connectors supplied directly to the government shall be marked with the military part number (see table I). Example: M28731/13-0074.
    b. Connectors supplied with contacts to government end users shall be marked with the military part number (see table I). Example: M28731/13-0074.
    c. Connectors supplied without contacts to government end users shall be marked as follows: Example: M28731/13F (screw or socket type from table I). The user shall complete the remaining portion of the military part number as shown in table I based upon contact type he assembles. Example: Ⓔ M28731/13F-0074.
    d. Polarization position marking is not required unless otherwise specified. When required polarization marking shall be in accordance with the connector polarization guide (see configuration A). The marking shall appear following the part number.
14. Applicable contact insertion and removal tools:

    Insertion MIL-I-81969/13-01
    Removal MIL-I-81969/11-01

15. When double wire crimp contact MIL-C-28731/23 is required for use with the connector, it shall be ordered separately.
16. Rectangular shields shall conform to MIL-C-28731/19-0007 through -0010, -0017 and -0018 must be ordered separately.
17. Recommended hardware (not furnished) for panel mounting: (4) .112-40-UNC-2A (length to suit) bolt with No. 4 washer and mating nut.

FIGURE 1. Connector, male rectangular - Continued.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Mating and unmating forces (torque): 100 inch-ounces, maximum.

Dielectric withstanding voltage: 1,250 V rms, 60 Hz, minimum.

Mechanical shock: High impact shock applicable.

Part number: See table I.

TABLE I. Part number information.

Military part number	Actuating hardware	Polarizing pin position		Contact type
		A .110 Dia.	B .090 Dia.	
M28731/13-0073	Jackscrew Type C	1	1	Crimp MIL-C-28731/20 1/
M28731/13-0074	Fixed socket Type F	1	1	Crimp MIL-C-28731/20 1/
M28731/13-0075	Jackscrew Type C	1	1	Solder hole MIL-C-28731/25 2/
M28731/13-0076	Fixed socket Type F	1	1	Solder hole MIL-C-28731/25 2/
M28731/13-0077	Fixed socket Type F	1	1	Wrappost MIL-C-28731/26 2/
M28731/13-0078	Fixed screw Type E	1	1	Crimp MIL-C-28731/20 1/
M28731/13-0079	Fixed screw Type E	1	1	Solder hole MIL-C-28731/25 2/
M28731/13-0080	Fixed screw Type E	1	1	Wrappost MIL-C-28731/26 2/

1/ See figure 1, note 11
2/ See figure 1, note 12

NOTE:
M28731/13-0073 supersedes M28731/13-0001 thru M28731/13-0036.
M28731/13-0074 supersedes M28731/13-0037 thru M28731/13-0072.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Preparing activity:
Navy - EC

(Project 5935-3488-10)

Review activities:
Army - AR, AT, MI
Navy - AS
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AL, AV, ME
Navy - MC, OS
Air Force - 19

Agent:
DLA - ES

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